CP-04 · MAINFRAME KNOWLEDGE RECOVERY

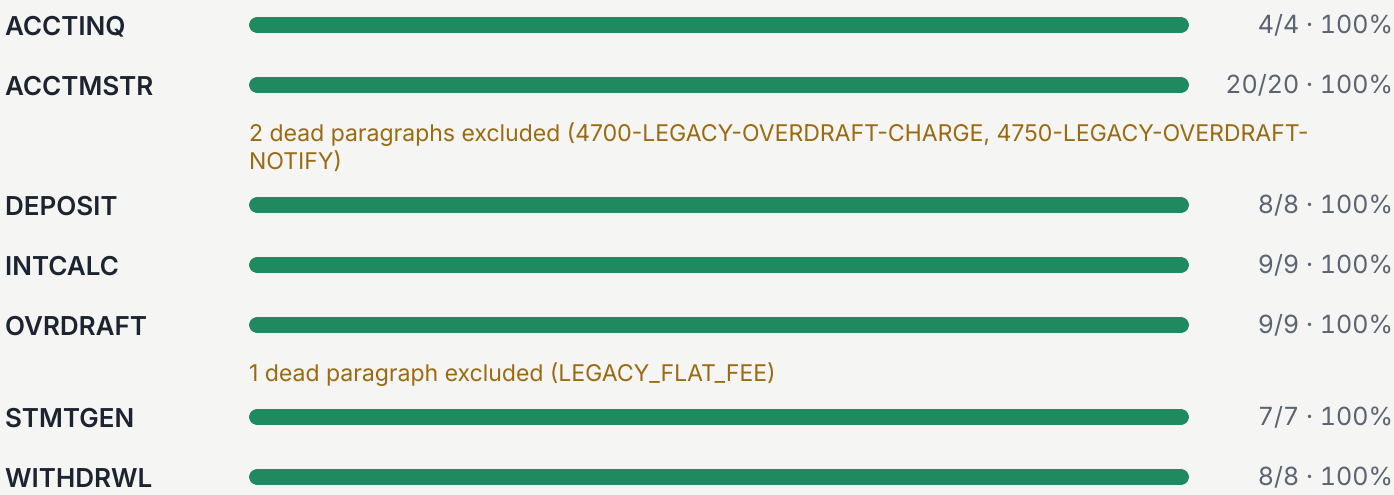
Rule Recovery Results

Live output of the extraction agent's run — across both COBOL and PL/I programs — every number on this page came out of `output/*.json`, none of it is hand-typed. Rules are unreviewed until the SME-confirm pill below turns green.

4	17	34%	0	1
PROGRAMS	RULES EXTRACTED	AVG. COVERAGE	DEAD-CODE LEAKS	DRIFT CAUGHT

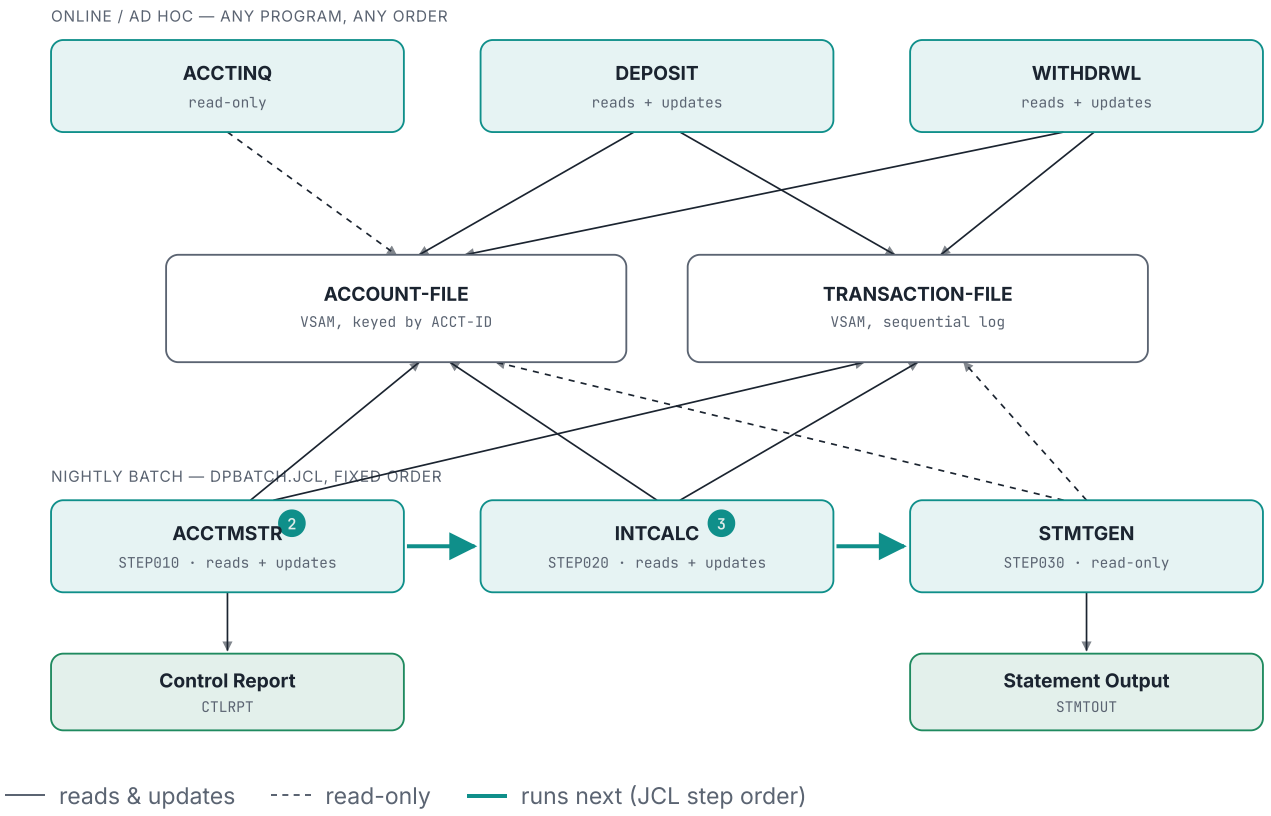
Coverage by program

Independently verified — counted from actual `get_paragraph_source` calls the agent made, not from what it claims.



How the six COBOL programs fit together

From the actual parsed file-access per program, not a guess — e.g. `STMTGEN` opens `ACCOUNT-FILE` read-only, so its "last-statement-date" update never persists to disk. (The PL/I program, `OVRDRAFT`, runs independently and isn't part of this shared-file picture — see its own card in "Inside each program" below.)

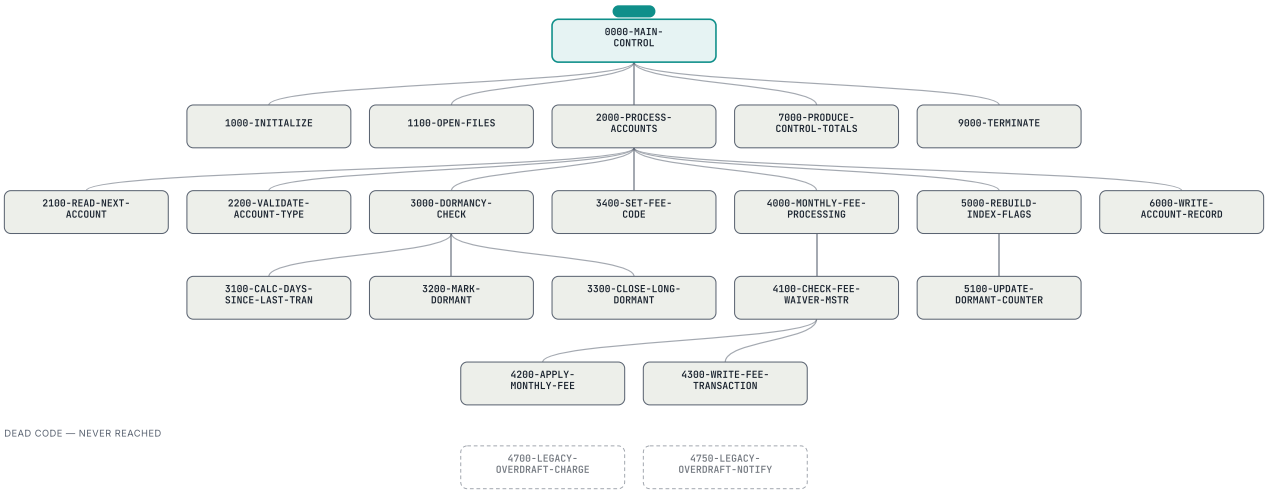


Final JCL step (STEP040, not drawn) unloads `ACCOUNT-FILE` / `TRANSACTION-FILE` into DB2 as `ACCOUNT_MASTER` / `TRANSACTION_LOG`.

Inside each program

The actual PERFORM graph, auto-laid-out by BFS distance from the entry paragraph — not hand-drawn. Hover a paragraph to trace what it calls. Dead paragraphs sit in their own row, never wired into the live flow.

ACCTINQ | ACCTMSTR · 2 dead | DEPOSIT | INTCALC | OVRDRAFT · 1 dead | STMTGEN | WITHDRWL



20 live paragraphs, 19 PERFORM edges, 2 excluded as dead code · entry: 0000-MAIN-CONTROL

Contradictions & drift caught

Two different checks: existing wiki pages diffed against extracted rules, and COBOL comparisons diffed directly against each other across programs.

FIELD RELATIONSHIP · ACCOUNT-MAINTENANCE-FIELD-REFERENCE.MD

Page describes `ACCT-INT-RATE` as independent. It physically shares storage with: `ACCT-FEE-CODE`, `ACCT-RATE-INFO`, `ACCT-RATE-INFO-R`.

PAGE SAYS
“ACCT-INT-RATE — the account's stored interest rate. Independent”

REDEFINES SAYS
Same physical bytes, 4 names

CROSS-PROGRAM THRESHOLD DRIFT

Same field, same literal value, compared with a different operator in two programs' live code:

ACCTMSTR
`ACCT-BALANCE > 500`

DEPOSIT
`ACCT-BALANCE ⩾ 500`

4100-CHECK-FEE-WAIVER-MSTR	3100-CHECK-FEE-WAIVER

NUMERIC DRIFT · ACCOUNT-MAINTENANCE-FIELD-REFERENCE.MD

PAGE SAYS	CODE SAYS (ACCTMSTR-A4)
365	180

Field lineage spotlight

The REDEFINES case — two field names sharing one physical storage slot.

`ACCT-FEE-CODE` = `ACCT-INT-RATE`

WRITTEN BY

`ACCT-FEE-CODE` ← **ACCTMSTR** 3400-SET-FEE-CODE

READ BY

`ACCT-INT-RATE` → **INTCALC** 2200-APPLY-RATE-TIER

DB2 MAPPING

`ACCT-FEE-CODE` → `ACCOUNT_MASTER.ACCT_FEE_CODE`

`ACCT-INT-RATE` → `ACCOUNT_MASTER.ACCT_INT_RATE`

EXCLUDED (DEAD CODE)

None

ACCT-INT-RATE physically shares storage (REDEFINES) with ACCT-FEE-CODE. They are the same bytes under different names, so a write to one is a write to all of them - lineage below covers the whole group. Written by: ACCTMSTR:3400-SET-FEE-CODE. Read by: INTCALC:2200-APPLY-RATE-TIER. Lands in DB2 as: ACCT-FEE-CODE → ACCOUNT_MASTER.ACCT_FEE_CODE; ACCT-INT-RATE → ACCOUNT_MASTER.ACCT_INT_RATE.

Dead code, correctly excluded

Reachability is computed from the PERFORM graph, not guessed — these paragraphs exist in source but were never given to the agent.

2 paragraph(s) in **ACCTMSTR** never reached from its entry point, so the agent never saw them: `4700-LEGACY-OVERDRAFT-CHARGE`, `4750-LEGACY-OVERDRAFT-NOTIFY`.

1 paragraph(s) in **OVRDRAFT** never reached from its entry point, so the agent never saw them: `LEGACY_FLAT_FEE`.

All extracted rules

Every row carries its exact source citation. None are authoritative until an SME confirms them.

Search rule text or paragraph name...

All (50) | ACCTINQ (3) | ACCTMSTR (13) | DEPOSIT (10) | INTCALC (8) | OVRDRAFT (7) | STMTGEN (4) | WITHDRWL (5)

ACCTINQ 3 rules · 100% coverage

`src/ACCTINQ.cbl:1000-READ-ACCOUNT` NEEDS SME CONFIRM

When a customer account inquiry is requested, the system looks up the account record; if no matching account is found, it displays an "ACCOUNT NOT FOUND" message and does not display any account details.

`src/ACCTINQ.cbl:1000-READ-ACCOUNT` NEEDS SME CONFIRM

This account inquiry function is read-only: it only displays the current balance and status, and never updates account information.

`src/ACCTINQ.cbl:2000-DISPLAY-BALANCE` NEEDS SME CONFIRM

When the account is found, the inquiry result displayed to the user includes the account identifier, the current account balance, and the account status.

ACCTMSTR 13 rules · 100% coverage

`src/ACCTMSTR.cbl:2100-READ-NEXT-ACCOUNT` NEEDS SME CONFIRM

Only Savings and Checking are recognized account types; any other account type is logged as a data exception but the account still continues through processing.

`src/ACCTMSTR.cbl:2200-VALIDATE-ACCOUNT-TYPE` NEEDS SME CONFIRM

When an account's type is neither Savings nor Checking, the system displays a data exception message identifying the account, but does not stop processing it.

`src/ACCTMSTR.cbl:3100-CALC-DAYS-SINCE-LAST-TRAN` NEEDS SME CONFIRM

The number of days since an account's last transaction is calculated by comparing today's date to the account's last transaction date.

`src/ACCTMSTR.cbl:3200-MARK-DORMANT` NEEDS SME CONFIRM

An active account with no transaction activity for more than 180 days is automatically reclassified as dormant.

`src/ACCTMSTR.cbl:3300-CLOSE-LONG-DORMANT` NEEDS SME CONFIRM

A dormant account with no transaction activity for more than 365 days is automatically closed by the system.

`src/ACCTMSTR.cbl:3400-SET-FEE-CODE` NEEDS SME CONFIRM

For checking accounts, a fee code is assigned based on balance: accounts with a negative balance get an overdraft fee code, while accounts with a zero or positive balance get the standard fee code.

src/ACCTMSTR.cbl:3400-SET-FEE-CODE

NEEDS SME CONFIRM

The fee code must only ever be set for checking accounts; savings accounts share the same stored data location for their interest rate, so setting a fee code on a savings account would corrupt its recorded interest rate.

---

src/ACCTMSTR.cbl:4100-CHECK-FEE-WAIVER-MSTR

NEEDS SME CONFIRM

The monthly maintenance fee is waived if the account balance is more than $500.00; if the balance is $500.00 or less, the fee is applied.

---

src/ACCTMSTR.cbl:4000-MONTHLY-FEE-PROCESSING

NEEDS SME CONFIRM

Monthly maintenance fee waiver checking is only performed for accounts that are currently active (not dormant or closed).

---

src/ACCTMSTR.cbl:4200-APPLY-MONTHLY-FEE

NEEDS SME CONFIRM

When a monthly maintenance fee is applied, it is deducted directly from the account's balance.

---

src/ACCTMSTR.cbl:4300-WRITE-FEE-TRANSACTION

NEEDS SME CONFIRM

Every time a monthly maintenance fee is applied, a fee transaction record is created showing the account, the fee amount, the resulting balance, and the date.

---

src/ACCTMSTR.cbl:5100-UPDATE-DORMANT-COUNTER

NEEDS SME CONFIRM

The account record keeps a running count of days since the last transaction, updated each processing cycle.

---

src/ACCTMSTR.cbl:7000-PRODUCE-CONTROL-TOTALS

NEEDS SME CONFIRM

At the end of each run, a control report is produced summarizing the total accounts processed, the number newly marked dormant, the number closed, and the number of monthly fees charged.

DEPOSIT 10 rules · 100% coverage

src/DEPOSIT.cbl:2000-READ-ACCOUNT

NEEDS SME CONFIRM

If the account number on the deposit cannot be found in the account file, the deposit is rejected.

---

src/DEPOSIT.cbl:2100-VALIDATE-ACCOUNT

NEEDS SME CONFIRM

A deposit is rejected if the account status is closed.

---

src/DEPOSIT.cbl:2100-VALIDATE-ACCOUNT

NEEDS SME CONFIRM

If the account is dormant (and not closed), a deposit automatically reactivates the account to active status.

---

src/DEPOSIT.cbl:3000-POST-DEPOSIT

NEEDS SME CONFIRM

The deposit amount is added to the account's current balance.

---

src/DEPOSIT.cbl:3000-POST-DEPOSIT

NEEDS SME CONFIRM

When a deposit is posted, the account's last-transaction date is updated to the current date.

`src/DEPOSIT.cbl:3100-CHECK-FEE-WAIVER` NEEDS SME CONFIRM

If the account balance after the deposit is at or above the fee-waiver threshold, the monthly maintenance fee is waived for this account.

`src/DEPOSIT.cbl:3100-CHECK-FEE-WAIVER` NEEDS SME CONFIRM

If the account balance after the deposit is below the fee-waiver threshold, a monthly maintenance fee is immediately deducted from the account balance.

`src/DEPOSIT.cbl:4000-WRITE-TRANSACTION` NEEDS SME CONFIRM

Every accepted deposit is recorded as a transaction log entry showing the account number, a transaction type of deposit, the deposit amount, the resulting account balance, and the date of the transaction.

`src/DEPOSIT.cbl:4000-WRITE-TRANSACTION` NEEDS SME CONFIRM

When a deposit is accepted, the account's stored record is updated to reflect the new balance and status.

`src/DEPOSIT.cbl:0000-MAIN` NEEDS SME CONFIRM

If the account fails validation (not found or closed), no posting, fee check, or transaction logging occurs for that deposit.

INTCALC 8 rules · 100% coverage

`src/INTCALC.cbl:2100-CHECK-ELIGIBLE` NEEDS SME CONFIRM

Interest is only calculated for accounts that are both a savings account type and currently active; checking accounts, dormant accounts, and closed accounts are not eligible to earn interest.

`src/INTCALC.cbl:2200-APPLY-RATE-TIER` NEEDS SME CONFIRM

If an eligible account's balance is $10,000.00 or more, the account earns a bonus annual interest rate of 4.5% instead of its normally stored interest rate.

`src/INTCALC.cbl:2200-APPLY-RATE-TIER` NEEDS SME CONFIRM

If an eligible account's balance is below $10,000.00, the account earns interest at its individually stored interest rate rather than the bonus rate.

`src/INTCALC.cbl:2300-COMPUTE-INTEREST` NEEDS SME CONFIRM

Monthly interest earned on an eligible account is calculated as the account's balance multiplied by its applicable annual interest rate, divided by 12, rounded to the nearest cent.

`src/INTCALC.cbl:3000-POST-INTEREST` NEEDS SME CONFIRM

The calculated monthly interest is added to the account's balance, crediting the account.

---

src/INTCALC.cbl:3000-POST-INTEREST

NEEDS SME CONFIRM

When interest is posted to an account, the account's last transaction date is updated to the current processing date.

---

src/INTCALC.cbl:4000-WRITE-TRANSACTION

NEEDS SME CONFIRM

Each interest payment is recorded as a transaction history entry showing the account number, a transaction type of 'IN' (interest), the interest amount credited, the resulting account balance after posting, and the date of posting.

---

src/INTCALC.cbl:4000-WRITE-TRANSACTION

NEEDS SME CONFIRM

The account's master record is updated to reflect the new balance and last transaction date whenever interest is posted.

OVRDRAFT 7 rules · 100% coverage

src/OVRDRAFT.pli:INIT_PROC

NEEDS SME CONFIRM

Every overdraft protection request is defaulted to "denied" and "no link" before processing begins, so that if processing fails partway through, the request is never mistakenly approved.

---

src/OVRDRAFT.pli:VALIDATE_LINK

NEEDS SME CONFIRM

If an account has no overdraft protection linked to it, the overdraft request is denied immediately without ever checking an overdraft limit.

---

src/OVRDRAFT.pli:CHECK_LIMIT

NEEDS SME CONFIRM

An overdraft request is approved only if the requested amount does not exceed the account's linked overdraft limit; otherwise it is denied.

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src/OVRDRAFT.pli:CALC_FEE

NEEDS SME CONFIRM

The first use of overdraft protection by an account in a calendar month is free of charge.

---

src/OVRDRAFT.pli:CALC_FEE

NEEDS SME CONFIRM

Every overdraft protection use after the first one in the same calendar month incurs a flat fee.

---

src/OVRDRAFT.pli:APPLY_FEE

NEEDS SME CONFIRM

Each time an overdraft fee is calculated and applied, the account's count of overdraft protection uses for the month is increased by one.

---

src/OVRDRAFT.pli:WRITE_RESPONSE

NEEDS SME CONFIRM

The response to an overdraft request always includes the approval decision and the fee amount, even when the fee is zero.

STMTGEN 4 rules · 100% coverage

src/STMTGEN.cbl:3000-PRINT-HEADER

NEEDS SME CONFIRM

Each customer statement begins with a header line showing the account identifier and the current account balance.

`src/STMTGEN.cbl:4000-PRINT-TRANSACTIONS` NEEDS SME CONFIRM

A statement lists only the transactions that belong to the account currently being statemented (i.e., transactions are matched to the account by account ID).

`src/STMTGEN.cbl:4000-PRINT-TRANSACTIONS` NEEDS SME CONFIRM

Each listed transaction line on the statement shows the transaction type, the transaction amount, and the transaction date.

`src/STMTGEN.cbl:5000-UPDATE-LAST-STMT-DATE` NEEDS SME CONFIRM

Whenever a statement is generated for an account, the account's last-activity date is updated to the current date, regardless of whether any customer transactions occurred during the statement period.

WITHDRWL 5 rules · 100% coverage

`src/WITHDRWL.cbl:2000-READ-ACCOUNT` NEEDS SME CONFIRM

A withdrawal request is rejected if the account cannot be found.

`src/WITHDRWL.cbl:2100-VALIDATE-STATUS` NEEDS SME CONFIRM

A withdrawal is rejected if the account is closed or dormant; a dormant account must be reactivated with a deposit before a withdrawal can be made.

`src/WITHDRWL.cbl:2200-VALIDATE-FUNDS` NEEDS SME CONFIRM

A withdrawal is rejected if completing it would drop the account balance below a required $100.00 overdraft cushion.

`src/WITHDRWL.cbl:3000-POST-WITHDRAWAL` NEEDS SME CONFIRM

When a withdrawal is approved, the withdrawal amount is deducted from the account balance and the account's last transaction date is updated to the current date.

`src/WITHDRWL.cbl:4000-WRITE-TRANSACTION` NEEDS SME CONFIRM

An approved withdrawal is recorded as a transaction record showing the account id, a transaction type of withdrawal, the withdrawal amount, the resulting account balance, and the transaction date, and the account's record is updated to reflect the new balance.